NOVA SCHOOL OF SCIENCE & TECHNOLOGY

Interação Pessoa-Máquina

2026/2027

Sous-Chef Ratchet: Kitchen Assistant

Stage 3: 1st Prototype



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Lab class Nº P1

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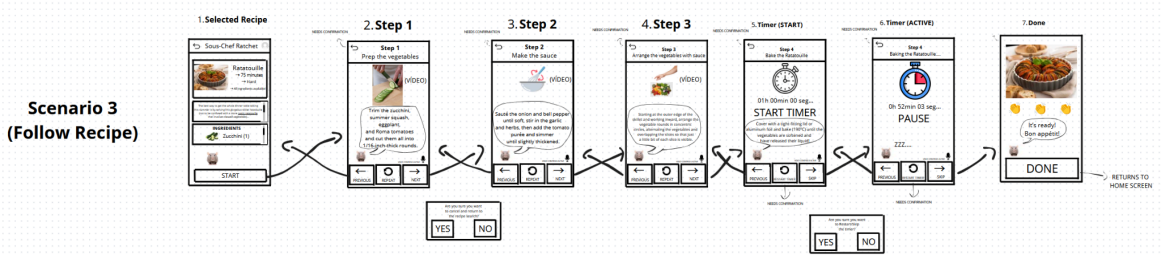
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Sketches

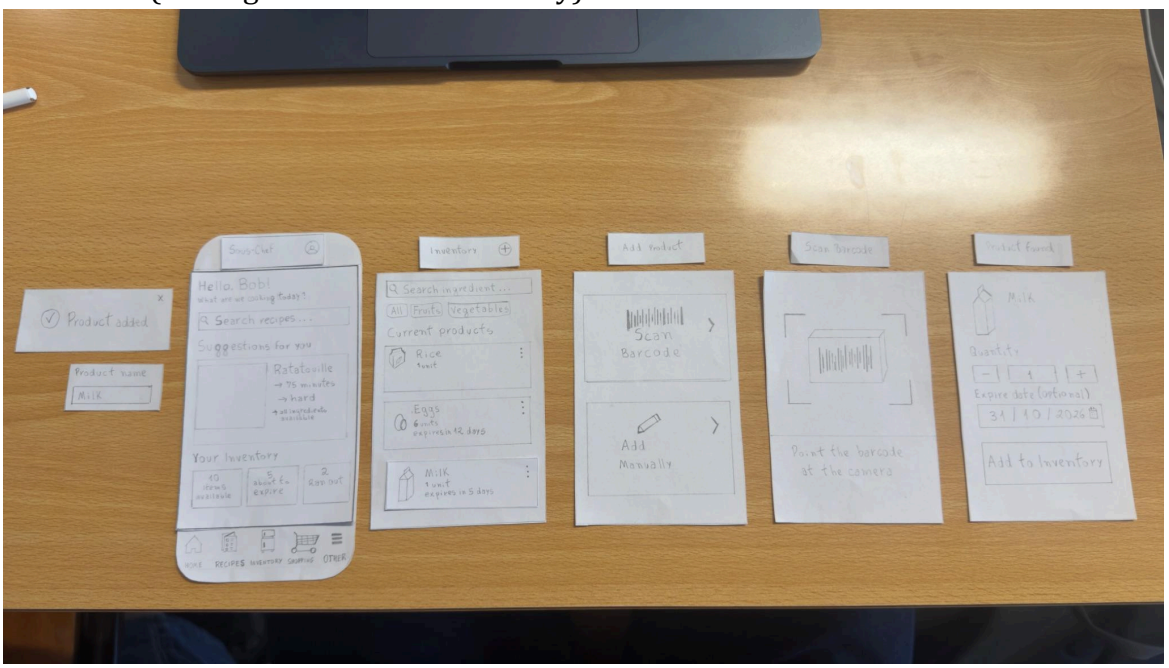
To keep the team coordinated and plan our drawings, we used Canva to draw the first versions of the prototype screens and keep them available to be seen at any time by any team member.
(Public viewing link: https://canva.link/ug3dvmvrdzukydy)

Example screenshot:

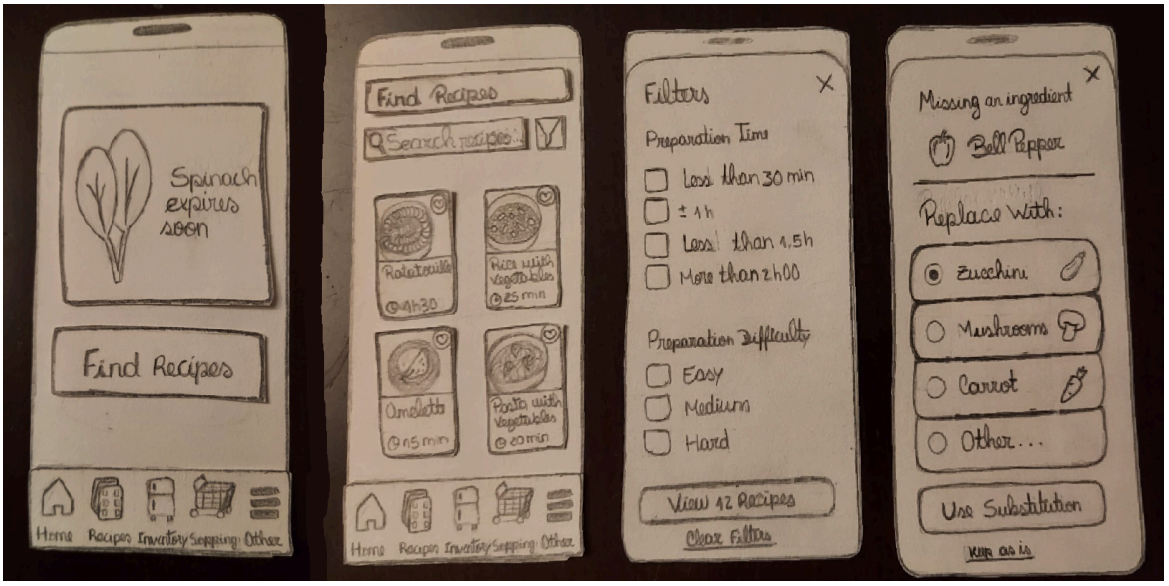


Prototype Photos

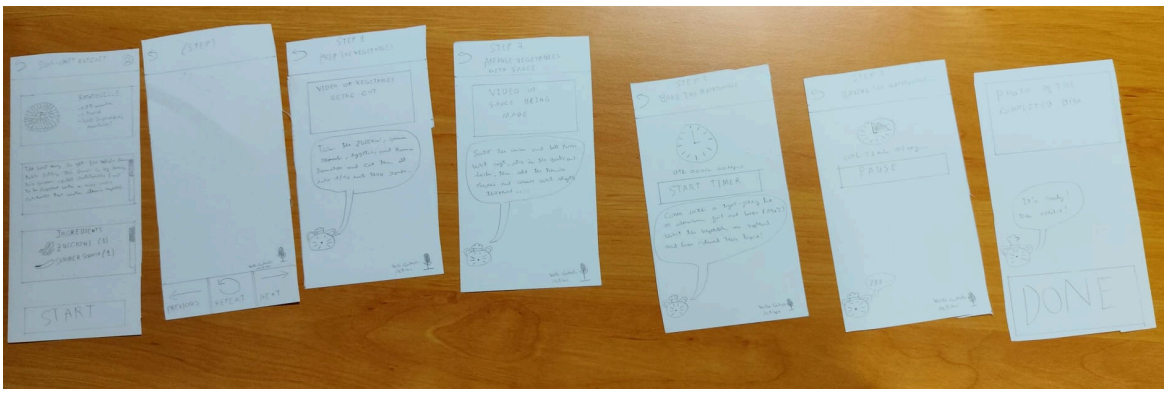
Scenario 1 (Adding Products to the Pantry)



Scenario 2 (Finding a Recipe with Available Ingredients)



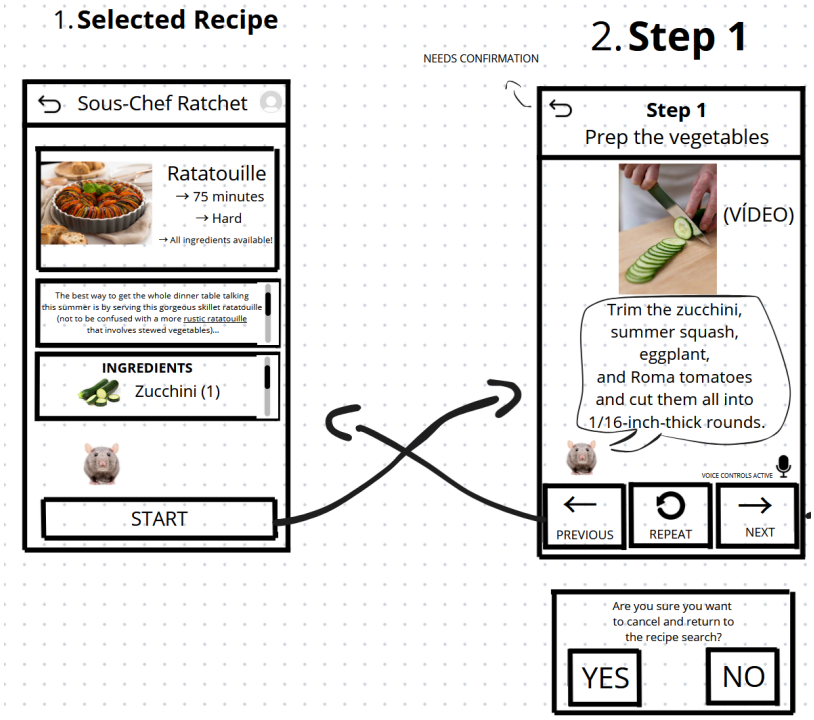
Scenario 3 (Following a Recipe When Cooking) Photos:



Storyboards

The Canva project mentioned earlier (https://canva.link/ug3dvmvrdzukydy) was also useful for planning the interactions between screens, and allowed us to prepare the storyboards there.

Example in Scenario 3, showing interactions between the first and second screen:



Briefing

“This is a mobile application that assists the user in kitchen/cooking related activities. The application allows you to manage your household inventory, organize your shopping list, search for recipes and get instructions while cooking.”

Scenarios

1. **Adding Products to the Pantry:**
   You share an apartment with two other people and often find that you buy ingredients you already have at home. Sometimes, you also discover that products have expired before you get a chance to use them.

   You have recently bought milk, eggs, and rice, and you want to keep track of what is available in your kitchen, including quantities and expiration dates where possible.

   Using the kitchen assistant app, add these products to your pantry and check that they have been recorded correctly.

2. **Finding a Recipe with Available Ingredients:**
   It is a weekday evening, and you have about 30 minutes to prepare dinner. You would prefer not to go shopping, as you already have several ingredients at home. Some vegetables in your refrigerator are approaching their expiration dates, and you would like to use them before they go to waste.

You are not sure what to cook, so you decide to use the kitchen assistant app to find a suitable meal. You want to choose something that fits your available ingredients, the time you have, and your cooking preferences.

Find a recipe you could realistically prepare tonight, taking into account any ingredients you may be missing.

3. **Following a recipe while cooking:**
   You have chosen a recipe in the kitchen assistant app and are ready to prepare dinner. You do not cook very often, so you need to refer to the instructions as you work.

   Your phone is on the kitchen counter, and you are preparing food with your hands. You want to follow the recipe without repeatedly stopping to clean your hands or losing track of your progress. During cooking, you need to check the next instruction, revisit an earlier step, repeat the explanation for a step, and set a timer for one of the cooking stages.

   Using the app, prepare the recipe while referring to the information you need throughout the cooking process.

Observation

For our first scenario, most users were mostly quick at understanding the interface, with the exception of one user that needed help understanding the context of the scenario better and a tip about which button to press in the inventory to add a product. We concluded that the button for adding products must be made more visible.

For our second scenario, showing the users multiple recipes when we had only planned one choice led to inevitably having to correct users to pick the planned recipe (Ratatouille). The users were also misguided by the scenario description mentioning the user only having 30 minutes to cook, while Ratatouille has a ~1h30 cooking duration.
Aside from this scenario description issue, there seemed to be no problems with the interface, and all users were able to navigate correctly without help.

Finally, for our third scenario, some users did not notice that voice controls were an option, and also did not try to start the timer correctly and instead tried skipping that step, although they were stopped by the pop-up confirmation and then understood the intended flow.
We conclude that voice control availability should be made more obvious and that the “Next” button during the timer phase should default to starting the timer instead of attempting to skip the step to make the design more intuitive.